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Enhancing high-solid anaerobic digestion of cattle manure using air nanobubble water: Performance improvements and microbial mechanisms under thermophilic and mesophilic conditions

Ruoyu Wu [a,b], Chengxiang Bai [c], Hailong Shi [d], Yujie Cao [e], Cheng Peng [a], Jiazi Niu [a], Jie Wang [a], Jianghe Chen [a], Qingfang Zhang [a,*]

[a] School of Metallurgy and Environment, Lanzhou University of Technology, Lanzhou, 730050, China
[b] Department of Municipal and Environmental Engineering, School of Eco-environmental & Chemical Engineering, Xi'an University of Technology, Xi'an, Shaanxi, 710048, China
[c] Gansu Yinguang Chemical Industry Group Co., Ltd, Baiyin, 730900, China
[d] Zhonghao Chengguang Chemical Research Institute Co., Ltd, 643200, China
[e] College of Dingxi Vocational and Technical, Dingxi, 730500, China

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ABSTRACT

High-solid anaerobic digestion (HSAD) is a promising strategy for cattle manure (CM) valorization, but its efficiency is often limited by slow hydrolysis, acid accumulation, and ammonia inhibition. This study investigated the effects of air nanobubble water (Air-NBW) on HSAD of CM under thermophilic (55 ± 1 °C) and mesophilic (37 ± 1 °C) conditions. Batch digesters were operated at CM-to-liquid mass ratios of 1:1, 1:2, and 1:4, with distilled water as the control. Air-NBW promoted soluble chemical oxygen demand (SCOD) release and volatile fatty acids (VFAs) formation, indicating enhanced hydrolysis and acidogenesis. Under thermophilic conditions, Air-NBW accelerated both soluble organic matter production and subsequent VFAs conversion, whereas under mesophilic conditions it mainly facilitated later-stage VFAs turnover after transient acid accumulation. Compared with the corresponding controls, cumulative methane yield increased by 23.1%, 10.2%, and 31.9% at thermophilic CM-to-liquid ratios of 1:1, 1:2, and 1:4, respectively, and by 21.4%, 15.7%, and 17.4% under mesophilic conditions. The modified Gompertz model further indicated improved methane production kinetics. Microbial analysis showed that Air-NBW was associated with the enrichment of hydrolytic bacteria and hydrogenotrophic methanogens, supporting the coordinated conversion of particulate organics to methane. Correlation analysis linked *Corynebacterium* with SCOD and *Methanosphaera* with methane production under thermophilic conditions, while mesophilic enhancement was more related to VFAs turnover and methanogenic recovery. These findings demonstrate that Air-NBW can improve HSAD performance and provide a practical strategy for enhancing renewable energy recovery from CM.

1. Introduction

The global intensification of livestock production has led to vast accumulations of manure, among which cattle manure (CM) predominates [1]. Improper handling and disposal of CM can cause air emissions, groundwater pollution, soil degradation, and the propagation of pathogenic microorganisms [2]. At present, direct land application remains the predominant CM management practice [3]; however, CM often harbors low levels of heavy metals—which, upon repeated soil application, can accumulate (e.g., Cd, Hg), and ultimately threaten crop safety [4]. Consequently, devising strategies for CM volume reduction, detoxification, and resource recovery has emerged as an urgent priority. Anaerobic digestion (AD) represents a leading manure-treatment technology that concurrently reduces waste, detoxifies contaminants, facilitates nutrient recovery, and produces renewable biogas [5]. Operating temperature and total solids (TS) content critically influence AD performance [6]. Mesophilic AD (25–40 °C) is widely implemented due to its operational stability [7], whereas thermophilic AD (> 45 °C), despite its higher energy demands and potential stability issues, offers faster hydrolysis rates, increased biogas yields, and enhanced methane

* Corresponding author.
E-mail address: zhangqf@lut.edu.cn (Q. Zhang).

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productivity [8]. Therefore, comparative evaluation of AD performance under mesophilic versus thermophilic conditions is essential. Total solids content affects reactor rheology, organic substrate hydrolysis, and inhibitor buildup, thereby exerting a pronounced impact on AD efficiency and biogas yields [9]. Conventional low-TS AD systems (4–7% TS) produce large volumes of liquid digestate, which raises handling costs and undermines environmental gains [4,10].

High-solid anaerobic digestion (HSAD), defined by TS contents above 8%, has emerged as a promising approach for livestock manure valorization, offering a compact footprint, high energy conversion efficiency, reduced digestate volumes, and elevated organic loading rates [11]. Nevertheless, HSAD is challenged by inherently slow hydrolysis kinetics, compromised reactor stability, and suboptimal methanogenic activity [12]. For example, the high lignocellulosic content of CM impedes enzymatic hydrolysis, causing reactor inhibition, extended lag phases, and stability deterioration, which together diminish methane yields and production rates [13]. Moreover, substrates with elevated TS levels (e.g., 25% TS agricultural residues) promote the accumulation of inhibitory intermediates, such as volatile fatty acids (VFAs) and total ammonia nitrogen (TAN), thus suppressing methane production [14]. Likewise, when digesting chicken manure at 9% TS, biogas productivity decreases from 0.32 to 0.13 $L{\cdot}g^{-1}$ VS as TAN concentrations rise from 3000 to 6000 $mg{\cdot}L^{-1}$ [15]. Accordingly, to enhance economic feasibility and operational robustness, it is imperative to integrate innovative strategies into HSAD, thereby enabling efficient CM valorization while mitigating environmental risks.

Nanobubble water (NBW) technology involves water saturated with nanobubbles [16], typically containing 10^6–10^8 bubbles per milliliter with diameters <1000 nm [17]. It exhibits unique properties distinct from ordinary water, such as high specific surface area, enhanced mass transfer efficiency, extended stability, elevated water molecule mobility, strong diffusivity, and free radical generation capability [18]. Consequently, this technology has found wide applications in fields like marine environmental restoration, water disinfection, and membrane fouling mitigation [19]. In recent years, NBW application in AD has gained traction due to the fundamental role of water in AD systems. Water not only accelerates substrate hydrolysis but also transports hydrolytic products and intermediates to microbial reaction sites [20], establishing the feasibility of NBW-AD coupling. Existing studies confirm that NBW generated with different gases enhances AD performance. For instance, in pig manure AD, NBW addition increased cumulative methane yield (CMY) to 192–225 $mL{\cdot}g^{-1}$ VS_{added}, representing a 19–39% improvement over the control group [21]. In waste activated sludge AD, N_2-NBW increased system Zeta potential by 17.1%. Higher Zeta potential enhances repulsive forces between sludge particles, dispersing flocs to improve microbial-substrate contact and substrate accessibility, ultimately boosting methane yield by 29.1% [22]. In food waste AD, N_2-NBW and Air-NBW increased hydrolysis-related enzyme activity by 8–47% in the first stage and methanogenesis-related coenzyme activity by 34%, benefiting both hydrolysis and methanogenesis phases [23]. Among these, Air-NBW is particularly suitable for practical engineering due to its simple preparation, low cost, and environmental benignity. However, research on Air-NBW-enhanced AD performance remains limited, particularly regarding mechanisms under high-solid conditions. This knowledge gap stems partly from the misconception that "air is incompatible with AD". In reality, controlled air introduction does not abort AD. On one hand, trace oxygen is rapidly consumed by aerobic/facultative bacteria, restoring anaerobic conditions promptly [24]. On the other hand, certain methanogens display oxygen tolerance. For example, *Methanosarcina*, *Methanocorpusculum*, and *Methanobrevibacter* can adapt to microaerophilic conditions by synthesizing antioxidant enzymes (e.g., superoxide dismutase, catalase, peroxidase) and adjusting their community structures in response to oxygen levels [25].

Given the technical challenges in HSAD for CM treatment, including low hydrolysis efficiency and poor system stability, combined with Air-NBW's proven potential to enhance AD performance and the unresolved knowledge gap in its mechanistic basis, this study aims to systematically evaluate the impact of Air-NBW on CM HSAD performance and disentangle its mechanisms in regulating hydrolysis kinetics, methane production, system stability, and microbial community dynamics. The study seeks to provide theoretical insights and technical guidance for optimizing CM HSAD, enhancing energy recovery efficiency, and maximizing environmental benefits.

2. Materials and methods

2.1. Materials

Fresh CM was collected from the Gansu Holstein Dairy Cattle Breeding Demonstration Center and stored at 4 ± 1 °C before the experiment. The anaerobic digested sludge used as inoculum was sourced from Gansu Yanchangbu Sewage Treatment Plant. Following collection, the sludge was acclimated under thermophilic (55 ± 1 °C) and mesophilic (37 ± 1 °C) conditions to serve as inoculum. Table 1 presents the key characteristics of thermophilic and mesophilic inocula, as well as CM, based on triplicate measurements.

2.2. Air-NBW

Air-NBW was prepared using a nanobubble generator (JWNP, AL01S, China). Air gas was introduced into the generator at a controlled pressure of 0.4 MPa to 0.5 MPa at a temperature of 20 ± 2 °C. DW was drawn into the generator's circulation system and cycled for 25 min to produce Air-NBW for use throughout the HSAD process in this study. The freshly prepared liquid appeared as white opaque micro-nanobubble water, which turned into a colorless transparent NBW after standing for 5 to 10 min.

2.3. Experimental setup

In this study, batch experiments were conducted using 1 L anaerobic fermenters. The sealed lid of the reactor was equipped with two sampling ports with an inner diameter of 6 mm. One port was connected to a gas collection bag via a silicone tube (inner diameter 6 mm) for biogas collection, while the other port enabled dynamic sampling of digestate samples through a peristaltic pump. The substrate used in the experiment was CM, and the inoculum was anaerobic digested sludge acclimated at the corresponding temperature (55 ± 1 °C/37 ± 1 °C). The total mass of substrate in each reactor was controlled at 800 g, with 80 g of inoculum sludge added. The experimental group used Air-NBW as the liquid medium, while the control group used distilled water (DW) as the liquid medium. Six operating conditions were established by adjusting the mass ratio of CM-to-liquid medium, i.e., 1:1 (TS = 19.4%), 1:2 (TS = 13.1%), and 1:4 (TS = 8.1%). Based on the combination of temperature variables (55 ± 1 °C/37 ± 1 °C) and medium types (Air-NBW/DW), each operating condition was systematically named. Under thermophilic conditions, the control groups with ratios of 1:1, 1:2, and 1:4 were

Table 1
Physicochemical characteristics of CM and inoculum.

Parameters	Unit	CM	Thermophilic inoculum	Mesophilic inoculum
TS	%	41.38 ± 0.1	6.43 ± 0.02	6.44 ± 0.01
VS	%	15.54 ± 0.12	2.46 ± 0.04	2.53 ± 0.05
VS/TS	%	37.55 ± 0.14	38.26 ± 0.12	39.28 ± 0.09
C	%	19.13 ± 0.23	21.94 ± 0.04	20.77 ± 0.05
N	%	0.99 ± 0.02	2.82 ± 0.04	2.83 ± 0.03
H	%	2.64 ± 0.03	3.56 ± 0.06	3.32 ± 0.04
S	%	0.37 ± 0.01	0.63 ± 0.05	0.58 ± 0.02
C/N	/	19.32	7.78	7.34
pH	/	8.16 ± 0.04	6.75 ± 0.02	7.07 ± 0.02

labeled as T1, T3, and T5, respectively, while the corresponding experimental groups were labeled as T2, T4, and T6. Under mesophilic conditions, the control groups with the corresponding ratios were labeled as M1, M3, and M5, respectively, and the corresponding experimental groups were labeled as M2, M4, and M6. Each operating condition was conducted in triplicate, resulting in a total of 36 reactors. To simulate actual engineering operating conditions, no artificial pH adjustment was performed during the experiment. The reactors were placed in constant-temperature water baths at 55 ± 1 °C and 37 ± 1 °C to maintain the reaction temperature, respectively. The reactors were manually shaken for 1 min at a fixed time daily to ensure sufficient contact between the inoculum and substrate.

2.4. Chemical analyses

Standard methods [26] were used to determine the TS, VS, total alkalinity (TA), pH, soluble chemical oxygen demand (SCOD), and TAN of the digestate samples. The free ammonia nitrogen (FAN) was calculated using a specific formula [27]. The concentration of VFAs was determined using a gas chromatograph (SP-7890 Plus, Ruihong, China) equipped with a high-performance organic acid column that is 30 m long, has an internal diameter of 320 μm, and a film thickness of 0.50 μm. This system is combined with a flame ionization detector, which efficiently separates and quantifies the VFAs by utilizing the differing interactions between the acids and the stationary phase of the column. The acids are then ionized in the flame to produce measurable signals, allowing for the detection of various VFAs. The volume and composition of biogas were determined using a gas flow meter (LML-2, Alpha, China) and a biogas analyzer (GA 5000, Geotach, United Kingdom), respectively.

2.5. Microbial community analysis

Digestate samples for microbial community analysis were collected at the early and final stages of AD, corresponding to day 3 and day 60, respectively. Before sampling, each reactor was thoroughly mixed to ensure sample representativeness. Samples were collected under aseptic conditions from triplicate reactors for each treatment and immediately stored at −80 °C before DNA extraction. The V3-V4 regions of bacterial DNA were amplified using primers 341F (CCTACGGGNGGCWGCAG) and 806R (GGACTACHVGGGTATCTAAT), while the V4-V5 regions of archaeal DNA were amplified with primers Arch 519F (CAGCMGCCGCGGTAA) and Arch 915R (GTGCTCCCCCGCCAATTCCT). DNA library sequencing was conducted on an Illumina PE2500 platform. All raw sequencing data are available at the National Center for Biotechnology Information (https://www.ncbi.nlm.nih.gov/). The accession numbers are PRJNA1118040 and PRJNA1118045. For the correlation analysis between environmental factors and microbial communities, microbial abundance data at the genus-level were matched with the physicochemical and performance parameters obtained from the corresponding reactors and sampling stages. The selected environmental and process variables included pH, SCOD, acetate, propionate, butyrate, valerate, total VFAs (TVFAs), TA, TVFAs/TA, TAN, FAN, CMY, and methane production rate.

2.6. Statistical analysis

Statistical analyses were performed using SPSS software version 27.0 for Windows. Significant differences in performance indicators were assessed using one-way analysis of variance (ANOVA) followed by Tukey's post hoc test. Methane yield data were fitted using a modified Gompertz model [28]. The relationships between environmental factors and microbial communities were analyzed using Pearson correlation analysis. Dominant microbial genera were selected for correlation analysis based on their relative abundance (RA) and ecological relevance. Before analysis, microbial RA data were transformed when necessary, and environmental variables were standardized to reduce the influence of different units and scales. Correlations with $P < 0.05$ were considered statistically significant, and those with $P < 0.01$ were considered highly significant. In the correlation networks, * and ** indicate $P < 0.05$ and $P < 0.01$, respectively.

3. Results and discussion

3.1. Digestion performance

3.1.1. pH

As a critical indicator of the HSAD process, pH exhibited similar dynamic trends in both thermophilic (Fig. 1a) and mesophilic (Fig. 1b) systems. All reactors experienced a pH drop to minimum values on day 3: in thermophilic groups, pH decreased from initial 7.7–8.2 to 7.2–7.6, whereas in mesophilic groups, it declined from 7.5 to 8.1 to 6.6–6.8. This was attributed to the rapid accumulation of VFAs during the hydrolysis and acidification phase. However, the extent of pH decrease differed markedly between the two temperature regimes. The smaller pH decline in the thermophilic reactors suggests that acidification was more effectively buffered under thermophilic conditions. This may be attributed to the faster release of ammonia nitrogen from protein degradation and the higher alkalinity generated during thermophilic digestion, which partly neutralized the acids produced during the early-stage [29,30]. In contrast, the sharper pH drop in the mesophilic reactors indicates that VFAs production exceeded their consumption by methanogens during the initial phase, leading to more evident transient acid stress. This interpretation is consistent with the subsequent VFAs and TVFAs/TA results.

As the reaction advanced, reactor pH rebounded and stabilized, driven by VFAs conversion and utilization alongside gradual increases in TAN concentration. Ultimately, thermophilic group pH stabilized at 8.1–8.2, while mesophilic group pH reached 7.5–7.8. The higher stabilized pH under thermophilic conditions favored acid-buffering capacity, but it also implies that FAN-related inhibition should be considered together with TAN and temperature, rather than judging stability from pH alone. Overall, the pH profiles suggest that Air-NBW-enhanced HSAD involved early acidification followed by recovery through VFAs conversion and alkalinity buffering. After stabilization, pH values in all reactors remained within the suitable range for methanogenic metabolism [31].

3.1.2. SCOD

As a primary indicator of the hydrolysis-acidogenesis stage in HSAD, SCOD reflects the extent of substrate disintegration and soluble organic matter release [32]. Fig. 2 shows that SCOD concentrations in all reactors increased initially and then declined. This behavior results from SCOD accumulation during hydrolysis and acidogenesis, followed by consumption by methanogens converting soluble organics into methane and carbon dioxide [33]. Under thermophilic conditions (Fig. 2a), SCOD peaked on day 3, reaching 10,724.3–32,922.7 $mg{\cdot}L^{-1}$, whereas mesophilic reactors (Fig. 2b) peaked later on day 8 at lower levels (5437.8–28,536.0 $mg{\cdot}L^{-1}$). The earlier SCOD peak under thermophilic conditions indicates that elevated temperature accelerated the solubilization of particulate organic matter from CM, while the subsequent rapid decline suggests faster conversion of soluble intermediates. In contrast, the delayed peak and slower decrease under mesophilic conditions indicate slower hydrolysis and downstream conversion at 37 °C, consistent with the stronger early pH decline and subsequent VFAs accumulation.

The effect of Air-NBW on SCOD depended on both temperature and solid-to-liquid ratio. Under thermophilic conditions, Air-NBW increased SCOD peaks in the 1:2 and 1:4 groups, especially in T6, where the peak was 54.0% higher than that in T5. However, T2 did not exceed T1 in the 1:1 groups, suggesting that excessive solid loading may mask the promoting effect of Air-NBW because SCOD accumulation is jointly

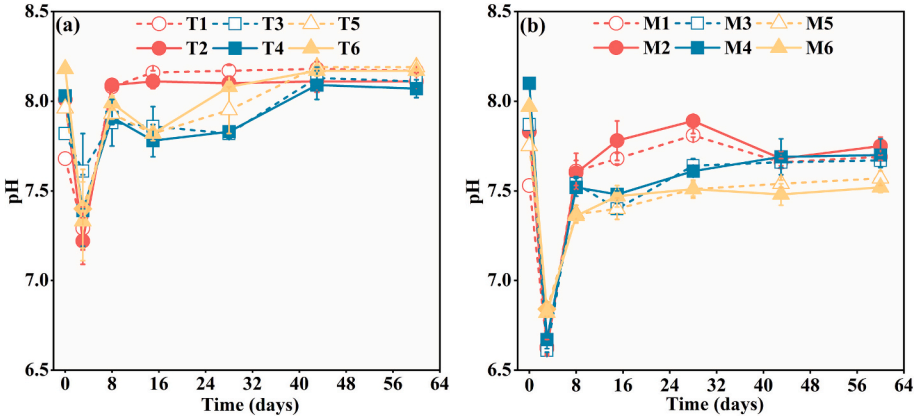


Fig. 1. Changes in pH under thermophilic (a) and mesophilic (b) conditions; dashed lines: control groups, solid lines: experimental groups.

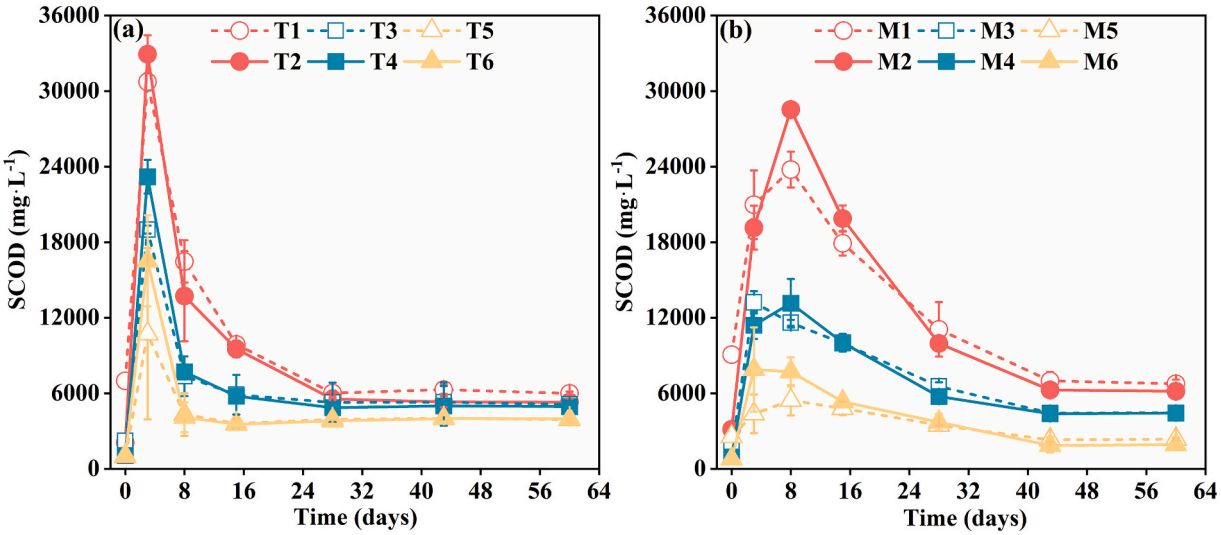


Fig. 2. Changes in SCOD under thermophilic (a) and mesophilic (b) conditions; dashed lines: control groups, solid lines: experimental groups.

controlled by substrate solubilization and soluble-organic conversion. Under mesophilic conditions, Air-NBW generally increased SCOD peaks, with M6 showing a 40.7% higher peak than M5, indicating enhanced soluble organic release. Nevertheless, the slower SCOD decline after the peak suggests that soluble intermediates were not rapidly converted at 37 °C. Overall, Air-NBW promoted SCOD release, but its effect was regulated by temperature and solid loading. Therefore, the enhanced methane production in Air-NBW reactors should be attributed not only to improved hydrolysis, but also to the subsequent conversion efficiency of soluble intermediates during methanogenesis.

3.1.3. VFAs concentration and components

VFAs are key intermediates linking hydrolysis-acidogenesis and methanogenesis in HSAD. They mainly consist of short-chain fatty acids, including acetic, propionic, and butyric acids [34]. Under thermophilic conditions, the VFAs concentrations in all groups reached their peaks on day 3 (Fig. 3). Among them, acetic acid is an important substrate for methanogenesis and is closely related to methanogenic conversion efficiency [35]. Air-NBW groups (T2, T4) exhibited acetic acid concentrations of 6132.7 and 3400.8 mg·L^{-1}, representing 46.3% and 66.1% increases over DW controls (T1, T3: 4193.2 and 2047.2 mg·L^{-1}), respectively. This suggests that Air-NBW promoted the formation of readily methanogenic intermediates under thermophilic conditions. After the peak, VFAs in thermophilic reactors decreased rapidly, and acetate was consumed earlier than propionate and butyrate. This pattern indicates that thermophilic digestion supported efficient acid conversion, shortened the period of acid accumulation, and improved methanogenic efficiency. Although T6 showed lower VFAs accumulation than some higher-solid groups, its methane performance was higher than that of T5, suggesting that Air-NBW may have improved not only intermediate production but also the subsequent utilization of VFAs under low-solid thermophilic conditions.

Under mesophilic conditions, high-solid loading groups (1:1, 1:2) (M1, M2, M3, M4) exhibited continuous VFAs accumulation for the first 15 days, reaching 15,056.4, 12985.2, 8378.7, and 9363.9 mg·L^{-1}, respectively, with minimal consumption. This indicates that, at 37 °C, hydrolysis and acidogenesis proceeded faster than methanogenic consumption during the early-stage, resulting in transient acid accumulation. This interpretation is consistent with the sharper pH decline and the later TVFAs/TA results. After day 15, the HSAD system gradually recovered, with rapid acetic acid consumption and gradual metabolism of propionic/butyric acid to methane within two weeks. Notably, Air-NBW groups (M2, M4) exhibited higher conversion efficiency, aligning with increased methane yields (see Section 3.1.6), confirming Air-NBW enhances VFAs consumption and methanogenic conversion, accelerating system stability recovery.

Overall, thermophilic digestion showed rapid VFAs production followed by rapid consumption, reflecting a better coupling between acidogenesis and methanogenesis. In contrast, mesophilic digestion showed delayed VFAs conversion and a longer acid stress period. Air-

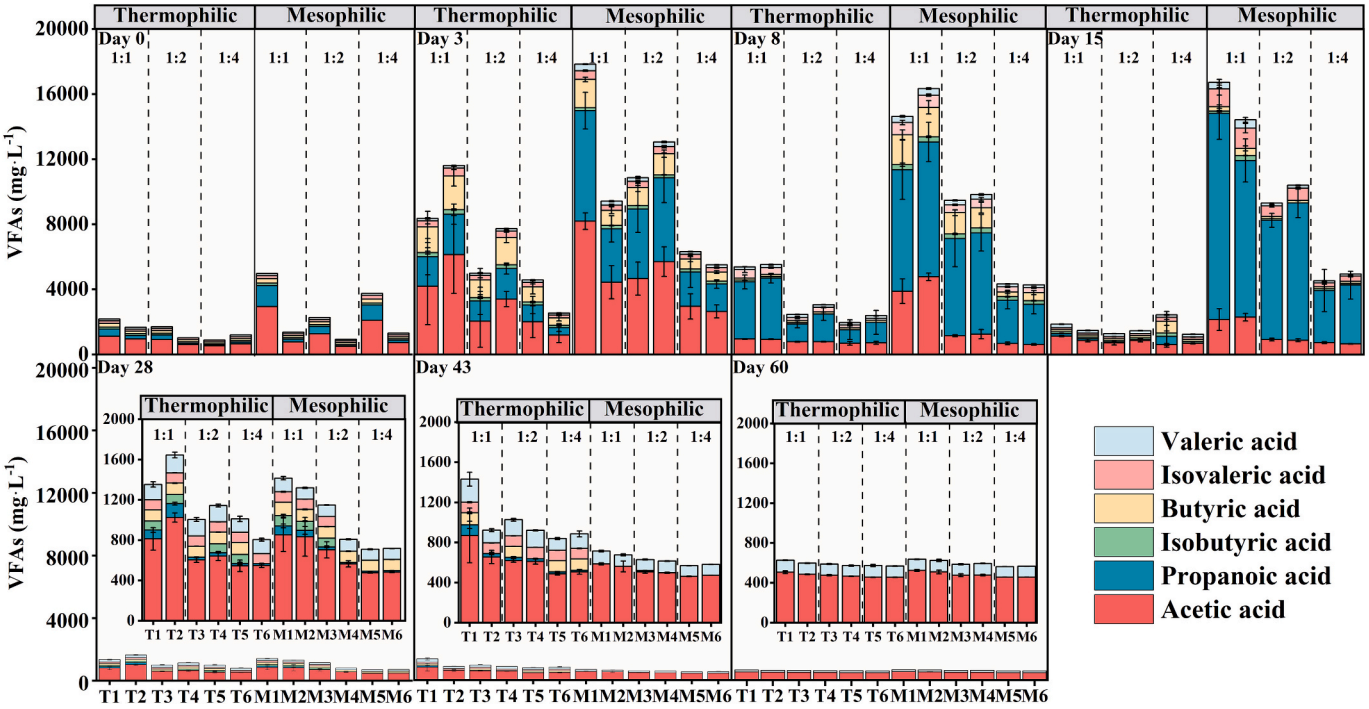


Fig. 3. Concentrations and compositions of VFAs under thermophilic and mesophilic conditions.

NBW promoted VFAs formation, especially acetate production, but its beneficial effect on methane production depended on whether the accumulated VFAs could be subsequently consumed. Thus, VFAs profiles indicate that Air-NBW-enhanced HSAD through both promoting intermediate formation and facilitating later-stage VFAs turnover, with the latter being particularly important under mesophilic conditions.

3.1.4. TA and VFAs/TA

TA, a determinant of HSAD buffering capacity, originates primarily from ammonia released during deamination of nitrogenous substrates [36]. Under both thermophilic and mesophilic regimes, TA in all digesters rose initially and then reached a plateau (Fig. 4a, b). This increase is attributable to the release of alkali metals (e.g., Ca^{2+}, Mg^{2+})

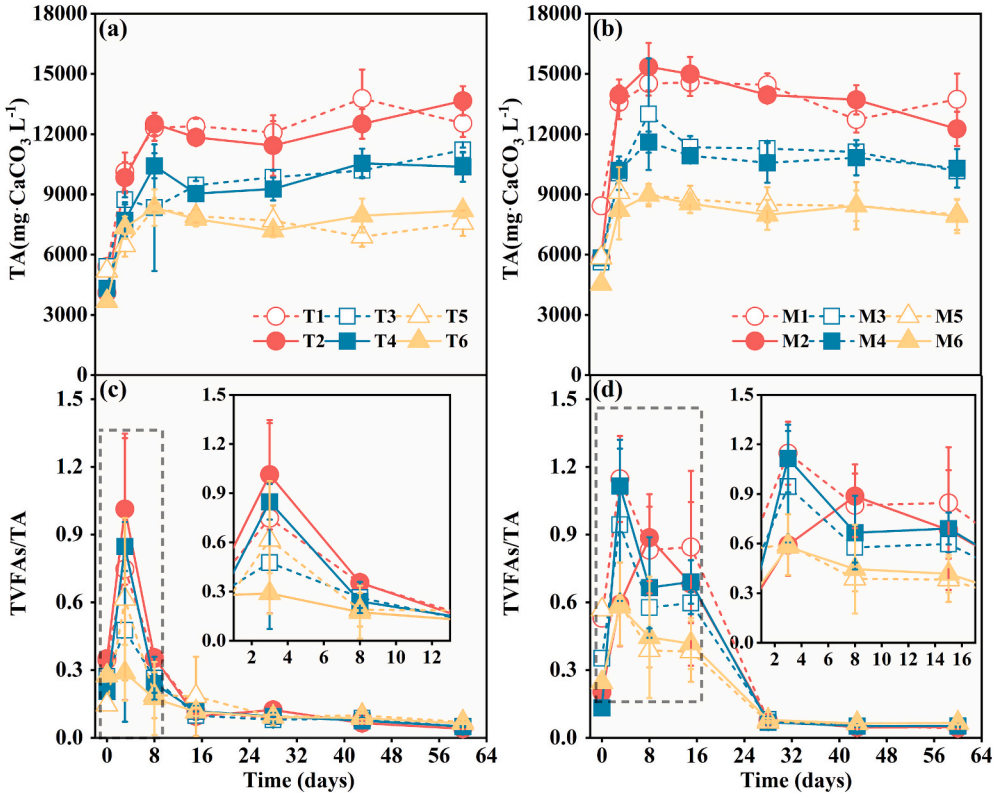


Fig. 4. Changes in TA and TVFAs/TA under thermophilic (a) (c) and mesophilic (b) (d) conditions; dashed lines: control groups, solid lines: experimental groups.

from CM, which contribute to TA during digestion [37]. Moreover, TA levels scaled with substrate solids content, since higher solid loadings correspond to increased nitrogenous organic matter and thus greater ammonia release through deamination. However, TA alone cannot fully represent process stability. Therefore, the TVFAs/TA ratio was further used to evaluate the balance between acid production and buffering capacity [38]. TVFAs/TA values below 0.35 denote adequate buffering and stable operation, whereas values above 0.8 indicate insufficient buffering capacity and potential process instability [4,39]. Under thermophilic conditions, TVFAs/TA reached a maximum on day 3 (Fig. 4c). Air-NBW treatments (T2, T4) exhibited TVFAs/TA ratios exceeding 0.8, indicating a transient destabilization due to enhanced hydrolysis. This transient instability resulted from accelerated VFAs accumulation driven by Air-NBW-enhanced hydrolysis. Thereafter, rapid VFAs consumption by methanogens drove TVFAs/TA below 0.35 by day 15, stabilizing the system in line with observed VFAs depletion (see Section 3.1.3). In mesophilic reactors, TVFAs/TA dynamics were delayed and fluctuated markedly during the initial 15 days (Fig. 4d). TVFAs/TA values in M1–M4 ranged from 0.6 to 1.1 well above the 0.35 threshold indicating acid inhibition and instability from VFAs accumulation during the first 15 days. As digestion continued, TVFAs/TA gradually declined below 0.35 by day 25, reflecting system recovery consistent with VFAs depletion trends.

Overall, TA results indicate that CM degradation provided buffering capacity, especially at higher solid loadings, whereas TVFAs/TA revealed whether this buffering capacity was sufficient to counteract acid accumulation. Thermophilic digestion showed a stronger ability to recover from early acid stress, while mesophilic digestion experienced a longer unstable period due to slower VFAs conversion. Air-NBW increased early acidification pressure by promoting hydrolysis, but it did not cause persistent instability; instead, its positive effect depended on whether VFAs conversion could subsequently keep pace with acid production.

3.1.5. TAN and FAN

TAN, a by product of protein hydrolysis, reflects digester hydrolysis capacity [40]. Moderate TAN concentrations support microbial growth by providing buffering capacity and nutrients. However, TAN concentrations >1500 mg·L^{-1} inhibit methanogen activity, reducing methane production [41]. In both thermophilic (Fig. 5a) and mesophilic (Fig. 5b) systems, TAN concentrations in high-solid (1:1) groups (T1, T2) stabilized at 1985.3–2014.5 mg·L^{-1}, while M1 and M2 reached 1734.5–1960.0 mg·L^{-1}, all exceeding 1500 mg·L^{-1} and indicating ammonia inhibition risk. In 1:2 and 1:4 ratio groups, TAN stabilized at 848.2–1482.2 mg·L^{-1} (T3, T4, T5, T6) and 799.0–1260.0 mg·L^{-1} (M3, M4, M5, M6), both below the inhibition threshold. Thus, the 1:2 and 1:4 digesters had a lower risk of TAN-related inhibition. Thermophilic groups exhibited higher TAN concentrations, likely due to enhanced hydrolytic activity in thermophilic microbial communities, highlighting hydrolysis limitations under mesophilic conditions [42].

FAN, a fraction of TAN, is more toxic to methanogens than TAN because it can penetrate microbial cell membranes and accumulate intracellularly, causing proton imbalance or inactivation of cytoplasmic enzymes [43]. When FAN concentration exceeds 250 mg·L^{-1}, it will inhibit methanogens in HSAD systems [44]. In both thermophilic (Fig. 5c) and mesophilic (Fig. 5d) groups, FAN concentrations increased rapidly after day 3, and FAN concentrations in thermophilic groups were generally higher than those in mesophilic groups, which is due to the positive correlation between FAN concentration and pH as well as temperature [45]. Among them, the FAN concentrations of thermophilic groups T1 and T2 finally stabilized at 289.4–320.3 mg·L^{-1}, exceeding 250 mg·L^{-1}, indicating that ammonia inhibition may have occurred in digesters with 1:1 ratio under thermophilic conditions. Notably, T2 exhibited lower FAN concentrations than T1, suggesting that Air-NBW may have partially reduced FAN-related stress under thermophilic 1:1 conditions. Remaining thermophilic groups had stable FAN concentrations of 134.9–210.1 mg·L^{-1}, while mesophilic groups showed 32.7–131.8 mg·L^{-1}, all below the inhibition threshold and indicating no significant ammonia stress.

3.1.6. Methane yield

Under thermophilic conditions, CMY and specific methane production rates in Air-NBW treatments (T2, T4, T6) significantly exceeded those of the DW controls (T1, T3, T5) (Fig. 6a and b). Specifically, the

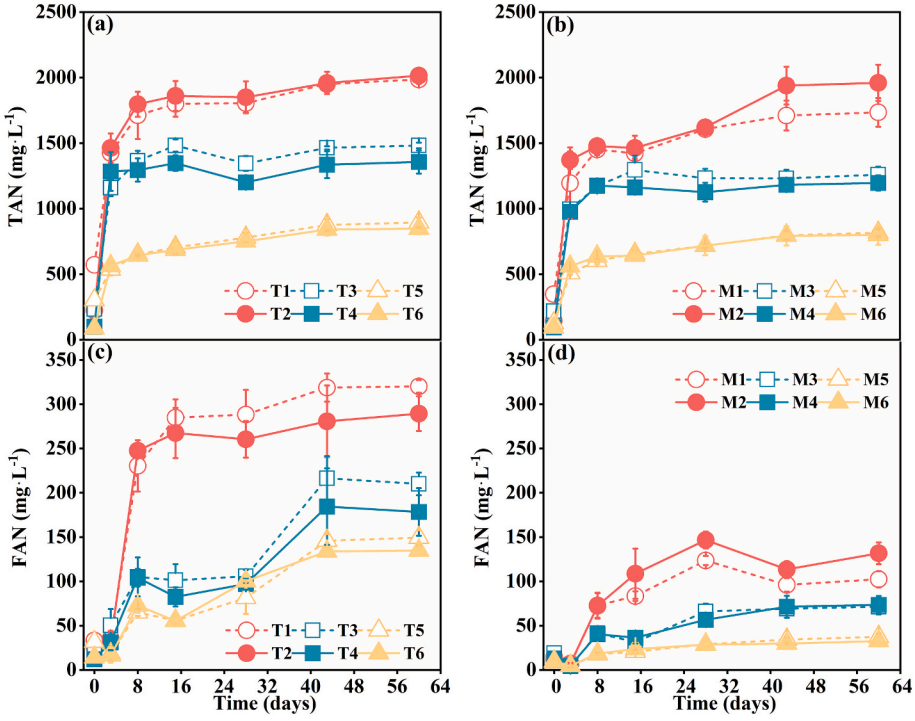


Fig. 5. Changes in TAN and FAN under thermophilic (a) (c) and mesophilic (b) (d) conditions; dashed lines: control groups, solid lines: experimental groups.

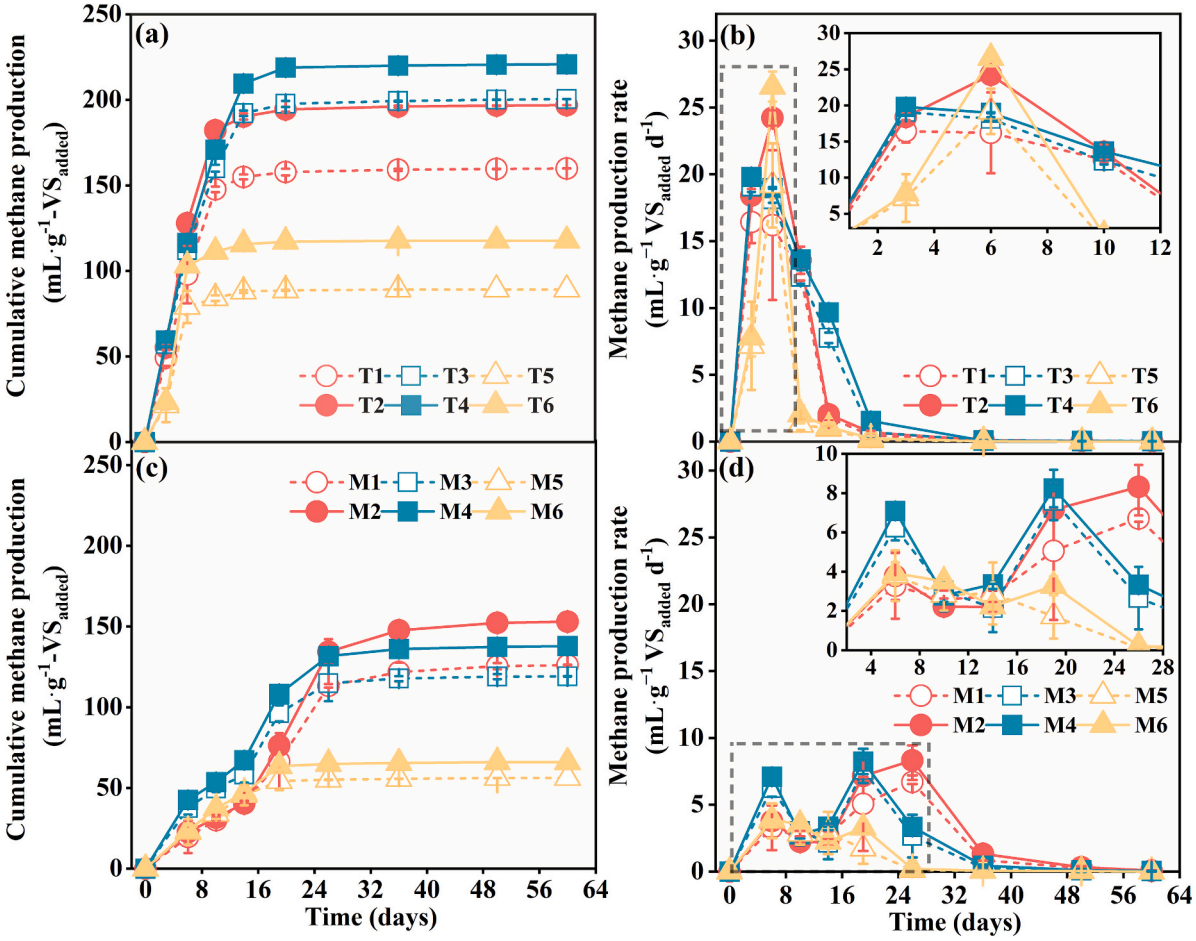


Fig. 6. Changes in CMY and methane production rate under thermophilic (a) (b) and mesophilic (c) (d) conditions; dashed lines: control groups, solid lines: experimental groups.

CMY values of T1-T6 were 159.9, 196.8, 200.4, 220.8, 89.2, and 117.6 mL·g^{-1} VS_{added}, respectively. Compared with the corresponding DW controls, Air-NBW increased CMY by 23.1%, 10.2%, and 31.9% in T2, T4, and T6, respectively. Although T1 and T2 had the highest TS and substrate loads and thus had the greatest theoretical methane potential, their CMY fell below those of T3 and T4 due to ammonia inhibition at excessive solids concentrations, as evidenced by TAN and FAN levels exceeding inhibitory thresholds (see Section 3.1.5). In T2, Air-NBW reduced FAN relative to T1, partially relieving ammonia toxicity and thereby contributing to the 23.1% CMY increase. Moreover, T2 achieved a specific methane production rate of 24.2 mL·g^{-1} VS_{added} d^{-1} versus 16.4 mL·g^{-1} VS_{added} d^{-1} in T1, indicating enhanced conversion efficiency (Fig. 6b). T3 and T4 recorded the highest absolute CMY values, yet the relative CMY enhancement in T4 was only 10.2%, and its specific methane rate of 19.8 mL·g^{-1} VS_{added} d^{-1} only marginally exceeded T3's 19.1 mL·g^{-1} VS_{added} d^{-1}. T5 and T6, with the lowest solids content and substrate load, exhibited the lowest CMY values; however, T6 displayed the greatest relative CMY increase (31.9%) upon Air-NBW addition and achieved a superior specific methane rate of 26.6 mL·g^{-1} VS_{added} d^{-1} versus 19.2 mL·g^{-1} VS_{added} d^{-1} in T5.

Under mesophilic conditions, the CMY values for M1–M6 were 126.0, 153.0, 119.1, 137.8, 56.1, and 65.9 mL·g^{-1} VS_{added}, respectively (Fig. 6c). Air-NBW treatments (M2, M4, M6) enhanced CMY by 21.4%, 15.7%, and 17.4% compared to their DW controls (M1, M3, M5), respectively. Although M1 had the highest TS and substrate load and thus the greatest theoretical methane potential, its actual CMY was lower than that of M4, likely due to acid inhibition from excessive VFAs accumulation. In contrast, M2, which shared the same solids content as M1 but was supplemented with Air-NBW, achieved a 21.4% CMY increase, and its specific methane production rate reached 8.3 mL·g^{-1} VS_{added} d^{-1} versus 6.7 mL·g^{-1} VS_{added} d^{-1} in M1 (Fig. 6d), suggesting that Air-NBW may have facilitated recovery from acid inhibition. M4 recorded the highest absolute CMY; however, its relative enhancement versus M3 was modest (15.7%), and its specific methane rate (8.2 mL·g^{-1} VS_{added} d^{-1}) only marginally exceeded M3's 7.6 mL·g^{-1} VS_{added} d^{-1}. M5 and M6, characterized by the lowest solids content and substrate load, exhibited the lowest CMY values; nevertheless, Air-NBW addition increased CMY in M6 by 17.4%, and M6's specific methane rate of 3.9 mL·g^{-1} VS_{added} d^{-1} slightly exceeded M5's 3.8 mL·g^{-1} VS_{added} d^{-1}, indicating improved substrate conversion efficiency. The modified Gompertz model further indicated improved methane production kinetics (Table 2). The experimental data fitted well with the model, with R^2 values ranging from 0.982 to 0.999. Compared with DW controls, Air-NBW increased both the modeled CMY and Rm under all corresponding conditions, indicating improved methane production potential and conversion kinetics. However, the slightly longer lag phase in Air-NBW treatments suggests that enhanced early hydrolysis and acidogenesis may have caused transient accumulation of intermediates before stable methanogenesis was established.

Overall, Air-NBW-enhanced methane production under all tested conditions, but the controlling mechanism differed between thermophilic and mesophilic digestion. Under thermophilic conditions, Air-NBW promoted rapid hydrolysis and methanogenic conversion, and the 1:2 ratio achieved the highest CMY by balancing substrate supply and inhibition risk. Under mesophilic conditions, Air-NBW improved methane yield mainly by facilitating the later conversion of accumulated

Table 2
Parameters of the modified Gompertz model.

Parameters	Cumulative methane yield, CMY ($mL \cdot g^{-1}$ VS_{added})	Maximum methane production rate, R_m ($mL \cdot g^{-1}$ VS_{added} d^{-1})	Lag phase, λ(d)	R^2
T1	159.59	16.55	0.79	0.997
T2	195.94	24.68	1.08	0.999
T3	199.94	18.97	0.50	0.997
T4	221.00	19.59	0.55	0.997
T5	88.13	19.21	2.22	0.998
T6	116.23	26.42	2.38	0.998
M1	129.90	6.44	5.13	0.982
M2	157.78	8.20	6.41	0.989
M3	121.65	7.59	1.05	0.987
M4	140.75	8.35	1.19	0.989
M5	56.22	3.78	1.04	0.995
M6	66.40	3.99	1.17	0.991

soluble intermediates, while the highest CMY occurred in the 1:1 Air-NBW group despite its greater acid inhibition risk.

3.2. Microbial community

3.2.1. Thermophilic microbial community

Bacterial community succession under thermophilic conditions is presented in Fig. 7a. At day 3 of thermophilic AD, four genera predominated (RA > 3%): *Solibacillus* (8.6–41.8%), *Defluviitalea* (15.1–38.8%), *Lysinibacillus* (3.7–17.8%), and *Corynebacterium* (2.7–16.6%), indicating that hydrolytic and fermentative bacteria were active during the early-stage of thermophilic HSAD.

Solibacillus, a key hydrolytic genus, secretes extracellular enzymes that accelerate polymer breakdown and produces VFAs to sustain methanogens [46]; this syntrophic interaction supports overall system stability and performance. In Air-NBW treatments, *Solibacillus* RA increased to 41.8% in T2, 29.6% in T4, and 41.8% in T6, compared to 8.6%, 23.7%, and 25.7% in the corresponding controls (T1, T3, T5). This enrichment corresponds to the heightened SCOD peaks (10,724.3–32,922.7 $mg \cdot L^{-1}$) and accelerated hydrolysis (peak at day 3) observed in Section 3.1.2, confirming that Air-NBW enhances hydrolytic efficiency via selective bacterial enrichment. *Defluviitalea*, a thermophilic genus enriched at elevated temperatures, contributes to lactate fermentation, polysaccharide degradation, and proteolysis [47]; its higher RA in T6 versus T5 aligns with T6's superior substrate conversion performance. *Corynebacterium* efficiently degrades complex organics, including cellulose and starch, via diverse extracellular enzymes [48]; its RA increased with CM ratio (1:1 > 1:2 > 1:4), explaining elevated SCOD in high CM treatments (T1, T2). The coexistence of these hydrolytic and fermentative genera suggests that thermophilic digestion promoted the rapid conversion of particulate organic matter into soluble organics and VFAs. This microbial pattern provides a biological explanation for the early SCOD and VFAs peaks observed under thermophilic conditions. By day 60, the bacterial community structure changed markedly. The RA of early-stage hydrolytic genera, such as *Solibacillus* and *Defluviitalea*, decreased substantially to 1.8–11.8% and 0.6–1.0%, respectively, while the proportion of unclassified and low-abundance taxa increased accordingly. This shift indicates that readily biodegradable substrates were largely depleted during the later-stage of digestion, thereby weakening the ecological advantage of hydrolytic and acidogenic bacteria. This result was consistent with the continuous decline in SCOD and VFAs after their early peaks, suggesting that the bacterial community gradually shifted from active hydrolysis and acidogenesis toward a more substrate-limited state.

Archaeal community succession under thermophilic conditions is presented in Fig. 7b. At day 3 of thermophilic AD, four genera predominated (RA > 3%): *Methanosarcina* (31.4–54.6%), *Methanobrevibacter* (25.2–45.8%), *Methanothermobacter* (13.4–23.8%), and *Methanosphaera* (1.4–3.1%). In Air-NBW treatments, *Methanobrevibacter* RA increased to 45.8% (T2), 34.8% (T4), and 42.3% (T6) compared with 39.1%, 27.6%, and 25.2% in T1, T3, and T5, respectively. Similarly, *Methanothermobacter* and *Methanosphaera* were more abundant in Air-NBW groups, indicating that NBW addition favorably modulates the methanogenic community. Despite ammonia inhibition in T1 and T2 (see Section 3.1.5), T2 maintained higher *Methanobrevibacter* abundance than T1, suggesting that Air-NBW mitigates ammonia toxicity via community optimization, contributing to a higher CMY in T2 (196.8 $mL \cdot g^{-1}$ VS_{added}) versus T1 (159.9 $mL \cdot g^{-1}$ VS_{added}). By day 60, substrate depletion had led to reduced nutrient availability. Subsequently, more competitive, fast-growing taxa outcompeted others for these limited nutrients, decreasing the RA of less competitive genera. For example, *Methanobrevibacter* RA declined to 4.5–29.7%, while *Methanosarcina*

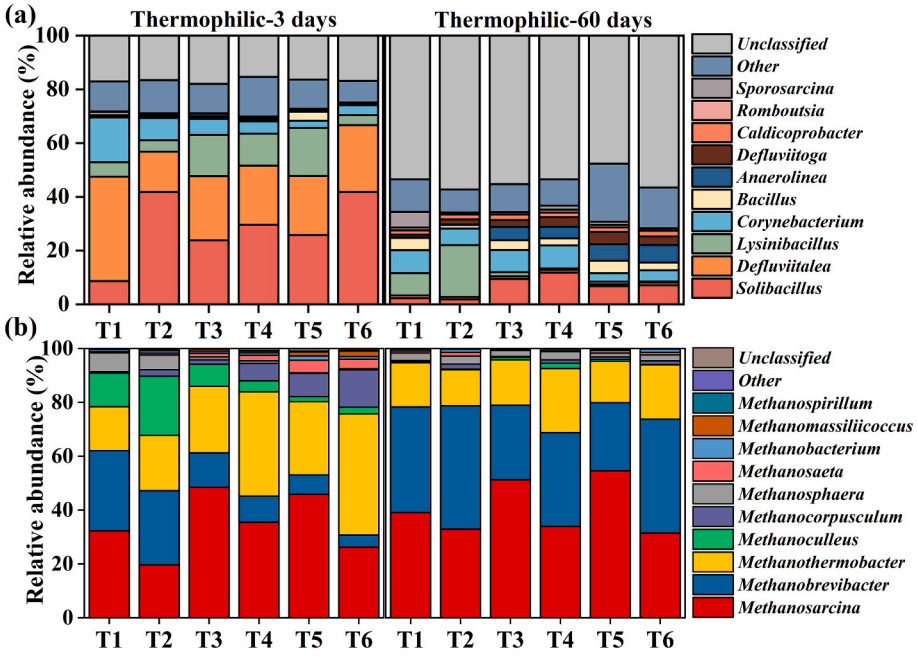


Fig. 7. Genus-level microbial community composition under thermophilic conditions: (a) bacterial community and (b) archaeal community.

declined to 19.6–48.4%.

3.2.2. Mesophilic microbial community

Bacterial community succession under mesophilic conditions is presented in Fig. 8a. At day 3, the bacterial community initially was dominated (RA > 3.0%) by *Solibacillus* (4.1–52.5%), *Lysinibacillus* (7.4–23.2%), *Acinetobacter* (1.1–19.5%), and *Corynebacterium* (0.9–8.5%). Compared with the thermophilic reactors, the mesophilic bacterial community maintained higher diversity, which may improve ecological resilience but also reflect a slower and less directional functional succession. This pattern was consistent with the delayed SCOD peak and prolonged VFAs accumulation observed under mesophilic conditions.

Solibacillus plays a key role in hydrolytic breakdown and stabilizes AD processes [46]. Its RA in the Air-NBW treatments (M2: 52.5%, M4: 9.1%) exceeded that in controls (M1: 40.7%, M3: 7.4%), mirroring the SCOD trends reported in Section 3.1.2. However, unlike the thermophilic system, the strengthened hydrolysis under mesophilic conditions did not immediately lead to efficient methane production. Instead, the produced soluble substrates and VFAs accumulated during the first 15 days, indicating that methanogenesis lagged behind hydrolysis and acidogenesis. *Acinetobacter* demonstrates versatile adaptability, thriving under both anaerobic and aerobic conditions. This versatility confers resilience to environmental fluctuations in AD, thereby supporting process stability [49]. Its RA rose with higher Air-NBW dosages (M6: 19.5% > M4: 2.4% > M2: 1.2%), potentially mitigating microenvironmental stress and accelerating recovery from acid inhibition (e.g., TVFAs/TA fell below 0.35 by day 25 in M6 versus M5). *Corynebacterium*, a known degrader of complex organics [48], correlated positively with valeric acid levels ($P < 0.05$), indicating that reactors enriched in *Corynebacterium* accumulated more organic acids, consistent with Fig. 3. By 60 day, bacterial diversity and dominant taxa declined markedly (e.g., *Solibacillus* to 0.7–21.2%), reflecting endpoint stabilization of SCOD and VFAs at low levels. Furthermore, mesophilic digesters exhibited higher microbial diversity than thermophilic ones throughout both early and late stages, in agreement with Chen et al.

Archaeal community succession under mesophilic conditions is presented in Fig. 8b. At day 3 of mesophilic AD, four genera predominated (RA > 5%): *Methanobrevibacter* (43.7–68.7%), *Methanosarcina* (10.1–27.1%), *Methanothermobacter* (2.5–6.9%), and *Methanocorpusculum* (3.6–5.3%). *Methanobrevibacter* and *Methanothermobacter* are hydrogenotrophic methanogens [25]. The high RA of *Methanobrevibacter* in mesophilic reactors indicates that hydrogenotrophic methanogenesis was an important pathway for methane production under mesophilic conditions. In particular, the RA of *Methanobrevibacter* was slightly higher in M2 and M6 than in their corresponding DW controls, increasing from 65.0% to 68.7% and from 43.7% to 44.9%, respectively. This suggests that Air-NBW may have favored the maintenance of hydrogenotrophic methanogens in some mesophilic treatments. Compared with acetoclastic methanogens, hydrogenotrophic methanogens are generally more tolerant to ammonia stress, which may help sustain methanogenic activity in nitrogen-rich CM digestion systems [50]. The enrichment of *Methanobrevibacter* in Air-NBW reactors was generally consistent with the improved methane yield observed in the corresponding treatments. However, because the increase in *Methanobrevibacter* abundance was not equally evident in all Air-NBW groups, the enhanced methane production under mesophilic conditions should not be attributed solely to this genus. Rather, it was likely associated with the combined effects of improved VFAs turnover, alleviated acid inhibition, and the recovery of methanogenic activity during the later-stage of digestion. Nevertheless, the overall methane production rate under mesophilic conditions remained lower than that under thermophilic conditions, indicating that Air-NBW improved methanogenic performance but could not completely overcome the kinetic limitation imposed by the lower operating temperature. *Methanosarcina*, which can perform both acetoclastic and hydrogenotrophic methanogenesis, was also detected in the mesophilic reactors [51]. However, its RA was generally lower than that in the thermophilic system, suggesting that the acetoclastic pathway may have been more vulnerable to VFAs accumulation and acid stress during mesophilic HSAD. This result helps explain why acetate accumulated for a longer period under mesophilic conditions, and why methane production was delayed despite the continuous release of soluble organic substrates.

3.3. Microbe-environment correlations

3.3.1. Thermophilic correlations

The correlation network between dominant microorganisms and environmental/process variables under thermophilic conditions is shown in Fig. 9a. The network revealed that the thermophilic HSAD

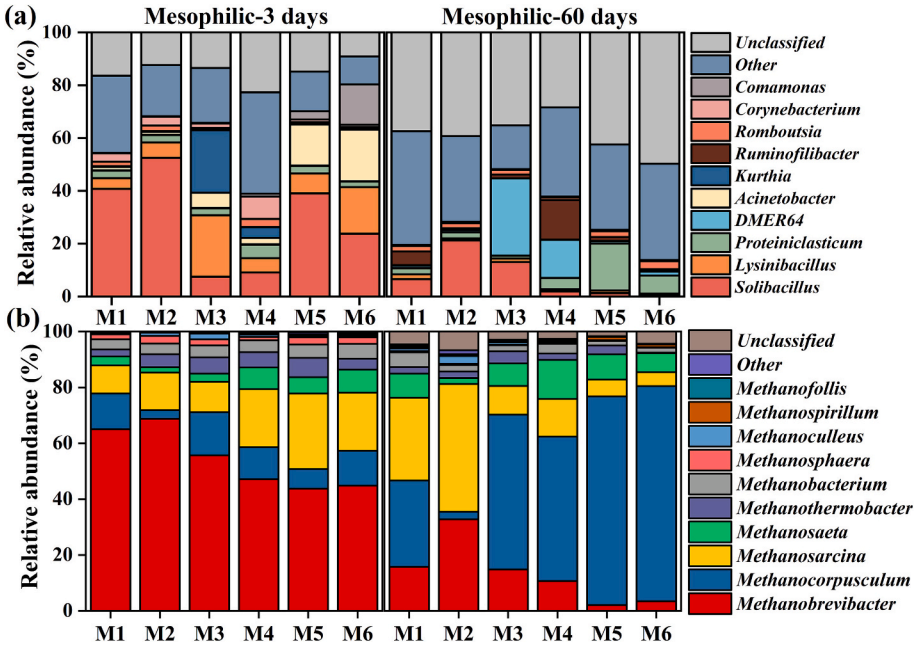


Fig. 8. Genus-level microbial community composition under mesophilic conditions: (a) bacterial community and (b) archaeal community.

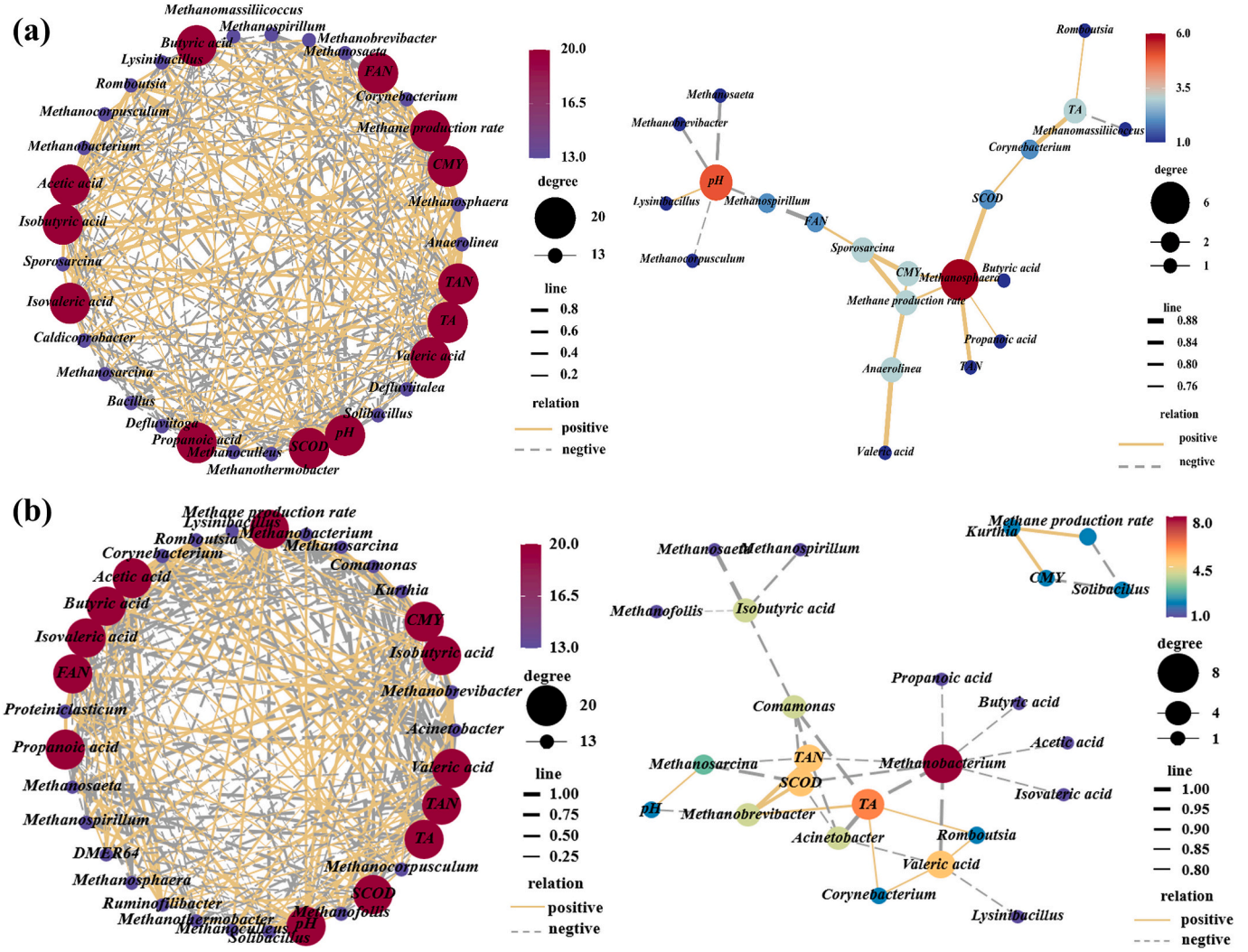


Fig. 9. Correlation networks between dominant microbial genera and environmental/process variables under (a) thermophilic and (b) mesophilic conditions.

process was regulated by the close coupling between hydrolysis-acidogenesis related variables and methanogenesis related variables. *Corynebacterium* was positively correlated with SCOD ($P < 0.05$), indicating that this genus was closely associated with the release of soluble organic matter during thermophilic digestion. This relationship supports the role of *Corynebacterium* in complex organic matter degradation and provides a microbial explanation for the rapid increase in SCOD during the early-stage. However, enhanced solubilization alone does not necessarily lead to higher methane recovery, because excessive soluble organics and VFAs may accumulate if methanogenic conversion is insufficient. Therefore, the relationship between hydrolytic bacteria and SCOD should be interpreted together with methanogenic indicators. In this study, *Methanosphaera* was positively correlated with both CMY and methane production rate ($P < 0.05$), suggesting that this archaeal genus was closely associated with efficient methane formation under thermophilic conditions. Although *Methanosphaera* was not the most dominant archaeal genus, its significant positive correlations with methane production indicators imply that it may serve as an important functional indicator of an active methanogenic community.

Therefore, the correlation results support a coordinated mechanism for Air-NBW-enhanced thermophilic HSAD. Air-NBW promoted the release of soluble organic matter by strengthening the hydrolytic bacterial process, while the thermophilic methanogenic community, represented by *Methanosphaera*-related methane production correlations, facilitated the downstream conversion of intermediates into methane. This microbial-process coupling explains why Air-NBW increased SCOD peaks without causing persistent instability and ultimately improved methane yield and methane production rate. Thus, under thermophilic conditions, the enhancement effect of Air-NBW should be understood as a coordinated improvement of hydrolysis and methanogenesis rather than as the enrichment of a single dominant taxon.

3.3.2. *Mesophilic correlations*

The correlation network between dominant microorganisms and environmental/process variables under mesophilic conditions is shown in Fig. 9b. The network revealed that mesophilic digestion was characterized by a weaker coupling between hydrolysis-acidogenesis and methanogenesis than thermophilic digestion. Hydrolytic and fermentative bacteria were closely associated with SCOD and VFAs related variables, whereas methanogenic archaea were more closely related to methane production and the depletion of fermentation intermediates. *Solibacillus* was negatively correlated with CMY and methane production rate ($P < 0.05$), indicating that enhanced hydrolysis and acidogenesis did not immediately translate into methane production under mesophilic conditions. This result is consistent with the prolonged VFAs accumulation during the first 15 days, suggesting an imbalance between acid production and acid consumption rather than a direct inhibitory effect of *Solibacillus* itself. *Corynebacterium* was positively correlated with valeric acid ($P < 0.05$), indicating that complex organic matter degradation and acidogenesis were still active in the mesophilic

reactors. This association supports the observation that VFAs accumulated more persistently under mesophilic conditions than under thermophilic conditions. These results indicate that the bacterial response to Air-NBW under mesophilic conditions was not simply beneficial or detrimental; rather, Air-NBW-enhanced hydrolysis and acidogenesis, while system performance depended on whether methanogens could consume the accumulated intermediates in time.

For archaeal communities, *Methanobrevibacter* was positively correlated with TAN ($P < 0.05$), suggesting that hydrogenotrophic methanogens played an important role in maintaining methanogenic activity under nitrogen-rich mesophilic conditions. This is particularly important for CM digestion, because protein degradation can continuously release ammonium nitrogen. The higher abundance of *Methanobrevibacter* in Air-NBW reactors indicates that Air-NBW may have improved the tolerance and functional stability of the methanogenic community under ammonia and acid stress. *Methanobacterium* was negatively correlated with acetate and propionate ($P < 0.05$), indicating that the increase in this genus was related to the consumption of VFAs or their upstream conversion products. This relationship supports the performance results showing that Air-NBW accelerated VFAs turnover during the later-stage of mesophilic digestion. Propionate is generally more difficult to degrade than acetate and is often associated with process instability when it accumulates. Therefore, the negative association between *Methanobacterium* and propionate suggests that hydrogenotrophic methanogenesis may have contributed to relieving acid inhibition and restoring digestion stability.

Overall, the correlation network suggests that Air-NBW-enhanced mesophilic HSAD through a mechanism different from that under thermophilic conditions. Under mesophilic conditions, Air-NBW promoted hydrolysis and acidogenesis, but the lower temperature limited the immediate conversion of VFAs to methane, leading to transient acid accumulation. Therefore, the subsequent performance improvement was mainly associated with accelerated VFAs turnover and the recovery of methanogenic activity, particularly hydrogenotrophic methanogens such as *Methanobrevibacter* and *Methanobacterium*, rather than simply with increased soluble organic matter release.

4. Conclusions

In this study, the effects of Air-NBW on the performance and microbial mechanisms of CM HSAD were investigated under thermophilic and mesophilic conditions. The main conclusions are as follows:

1. Air-NBW-enhanced hydrolysis and acidogenesis during HSAD of CM. This was reflected by increased SCOD release and VFAs formation. Thermophilic digestion showed faster hydrolysis and VFAs conversion, whereas mesophilic digestion exhibited delayed VFAs consumption and transient acid accumulation.
2. Air-NBW improved process stability by regulating acid accumulation and buffering recovery. Although Air-NBW increased early acidification pressure in some reactors, TVFAs/TA subsequently declined as VFAs were consumed and alkalinity increased. Thermophilic systems recovered more rapidly, while mesophilic systems experienced a longer acid-stress period.
3. Air-NBW improved methane production under all tested conditions. Compared with the corresponding DW controls, CMY increased by 10.2–31.9% under thermophilic conditions and by 15.7–21.4% under mesophilic conditions. The modified Gompertz model further indicated improved methane production kinetics.
4. Microbial and correlation analyses indicated that Air-NBW-enhanced HSAD through coordinated hydrolysis and methanogenesis. Thermophilic enhancement was associated with efficient organic solubilization and methanogenic conversion, while mesophilic enhancement relied more on VFAs turnover and methanogenic recovery.

CRediT authorship contribution statement

Ruoyu Wu: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation. **Chengxiang Bai:** Resources, Investigation. **Hailong Shi:** Resources, Investigation. **Yujie Cao:** Resources, Investigation. **Cheng Peng:** Resources, Investigation. **Jiazi Niu:** Resources, Investigation. **Jie Wang:** Resources, Investigation. **Jianghe Chen:** Resources, Investigation. **Qingfang Zhang:** Validation, Supervision, Methodology, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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